

Customer Behavior & Sales Insights

Project Summary

This project analyzes retail sales data to understand how customer segments, product categories, sub-categories, and discounting relate to overall sales and profitability.

I used **BigQuery SQL** to analyze business performance and **Tableau** to create an interactive dashboard highlighting the strongest patterns in the data.

The goal was to move beyond sales volume alone and identify areas where strong revenue was - and was not - translating into profitable business performance.

Business Question

What customer behaviors and product trends are driving sales performance, and where are the biggest opportunities for the business to improve profitability?

Tools

BigQuery SQL - Data analysis, aggregation, segmentation, and profitability calculations

Tableau - Data visualization and interactive dashboard development

Key Findings

Customer Segment Performance

The Consumer segment generated the highest sales at approximately **\$1.16 million** and the highest total profit at approximately **\$134,119**. However, Consumer had the lowest profit margin of the three customer segments at **11.55%**.

Home Office generated substantially less revenue but produced the highest segment profit margin at **14.03%**, followed by Corporate at **13.03%**.

This demonstrates that the segment generating the most revenue is not necessarily the strongest performer when profitability is considered.

Category Performance

Technology generated approximately **\$836,154 in sales** and achieved the highest category profit margin at **17.40%**.

Office Supplies performed similarly, producing a **17.04% profit margin**.

Furniture generated approximately **\$742,000 in sales**, but its profit margin was only **2.49%**. This indicates that significant Furniture revenue was not translating into comparable profitability.

Sub-Category Performance

Profitability varied considerably within individual product categories.

Copiers generated approximately **\$55,618 in profit** with a **37.20% profit margin**, while Paper achieved a **43.39% margin**.

Several products showed significantly weaker performance. Tables generated approximately **\$206,966 in sales but lost \$17,725**, resulting in a **-8.56% margin**.

Bookcases and Supplies also generated overall losses.

Machines generated approximately **\$189,239 in sales**, but only about **\$3,385 in profit**, resulting in a profit margin of just **1.79%**.

These results demonstrate why product performance should be evaluated using profitability alongside revenue.

Discount Impact

Discount levels showed a strong relationship with profitability in this dataset.

Transactions receiving **no discount produced a 29.51% profit margin**. At a **20% discount**, the overall margin remained positive at **11.82%**.

At a **30% discount**, however, profitability became negative at **-10.05%**.

Every analyzed discount level from **30% through 80%** generated an overall loss.

The results suggest that higher discount levels should be investigated as a potential area for improving profitability.

Recommendations

The business should evaluate performance using **both revenue and profitability**, rather than prioritizing sales volume alone.

Consumer customers remain important because of their scale, while Corporate and Home Office should be evaluated for opportunities to expand higher-margin business.

The company should continue supporting strong-performing categories such as Technology and Office Supplies while investigating the pricing, costs, and discount strategies affecting Furniture profitability.

At the product level, management should investigate underperforming sub-categories such as **Tables, Bookcases, Supplies, and Machines** to determine whether pricing, product costs, discounting, or product mix are limiting profitability.

The company should also review transactions receiving discounts of **30% or greater**. Additional analysis should determine whether discount level, product mix, customer segment, or another factor explains the negative profitability associated with these transactions before changes are made to discount policies.

Limitations

This analysis identifies patterns and associations within the available retail sales data but does not establish that any individual factor directly causes higher or lower profitability.

For example, higher discount levels are associated with lower profitability in this dataset, but the analysis does not establish that discounting alone caused the losses. Product costs, product mix, customer

behavior, shipping, and other business factors may also contribute.

The dataset is being used as a **portfolio business-analysis exercise** and should not be interpreted as representing the current performance of an actual company.

Additional analysis could examine sales trends over time, geographic performance, individual customer behavior, and interactions between product categories, customer segments, and discount levels.